

MODEL PROJECT FOR ESTABLISHMENT OF BREED MULTIPLICATION FARM

Introduction of farmer

The farmer or the firm have to provide complete information on himself or his firm here Name of the applicant: Current occupation/current business and area of work:

Details of past experience of farm operations especially dairy farm operations/details on present arrangement with veterinary consultant or any such firm if any:

Purpose of establishment of the farm:

Please mention motivation that led you to decide to start a breed multiplication farm/ why you wish to establish this farm. Specifically describe gaps (availability of quality animals/training for farmers and entrepreneurs/supply of quality milk/milk products) that you wish to bridge by establishing this farm in your area.

Information on already available infrastructure:

Please describe current land holding, current dairy farm infrastructure, milk processing and marketing arrangements ect. Also describe current viability of the farm. You may put some photographs for existing infrastructure.

PROJECT TITLE: -

Establishment of a breed multiplication farm with the capacity to manage 200 dairy cows and to meet out demand of milk in the area and supply quality heifers to entrepreneurs and farmers .

GOAL OF THE PROJECT: -

Breed multiplication farm is proposed to be established through entrepreneurship model for making available high genetic merit heifers (HGM) to farmers to fulfill their need of general shortage of such animals and also meet demand of the milk in the area and to augment the economy of rural livestock small holders, thereby improving the quality of life of poor farmers in the area.

OBJECTIVES: -

The main objectives of the farm are to:

- i) Establish dairy farm with the capacity to manage 200 dairy animals of indigenous breed/high yielding breed;

- ii) Multiplication of high yielding milch animals through scientific breeding including IVF technology and sex sorted semen
- iii) Spreading awareness about scientific management practices including animal nutrition, disease prevention etc.
- iv) Produce superior male and female calves for distribution in the field for genetic up-gradation of local cattle of the area;

TOTAL FINANCIAL OUTLAY: -

Amount of Rs 4.25 crores will be required for establishment of breed multiplication farm with the capacity of 200 milch animals.

TIME FRAME:-

Establishment of dairy farm will be completed in one year.

TECHNICAL PROGRAMME:-

- 200 cows/ buffalo of high yielding preferably of indigenous breeds will be inducted at the farm. Animal of indigenous -----breed will be purchased from the breeding tract. Indigenous animals will be yielding more than 3000 kgs per lactation (standard lactation yield). (meeting yield as mentioned in project document). Indigenous breed is selected as there is more demand of milk of indigenous breed in the area.
- Animals will be inseminated with sex sorted semen preferably of progeny tested bulls with dams lactation yield above 4000 kgs in case of indigenous cattle/ buffalo breeds. Semen straws will be purchased from different semen stations in the country.
- Farm will use IVF technology and implant embryos of indigenous cattle/buffalo breeds with high yielding potential. In embryo production donor of indigenous cattle/buffalo breeds may be yielding more than 4000kg and semen may be from indigenous cattle /buffalo bulls with dams lactation yield above (as per standards and specifications given in the guidelines). Bulls used in embryo production will be progeny tested/ genomically tested.
- About 140 female calves and 15 male calves will be born at the farm annually. 20 female calves from the elitist of elite dams will be retained at the farm for replacement of the herd @ 20% per annum. About 116 female calves would be sold to the farmers in the region.
- Animals will be tested annually for Tuberculosis, Johnes Disease (JD), and Brucellosis and positive animals will be removed from the herd. Herd

will be declared free of above diseases. The animals will be vaccinated against FMD (six month interval), HS (annually), BQ (annually), and Theileriosis (once in lifetime in case of exotic/crossbreds). However, the vaccination against the bacterial diseases will be done only if there is an outbreak or prevalence of a particular disease in the State. Exotic animals will also be tested for genetic disorders.

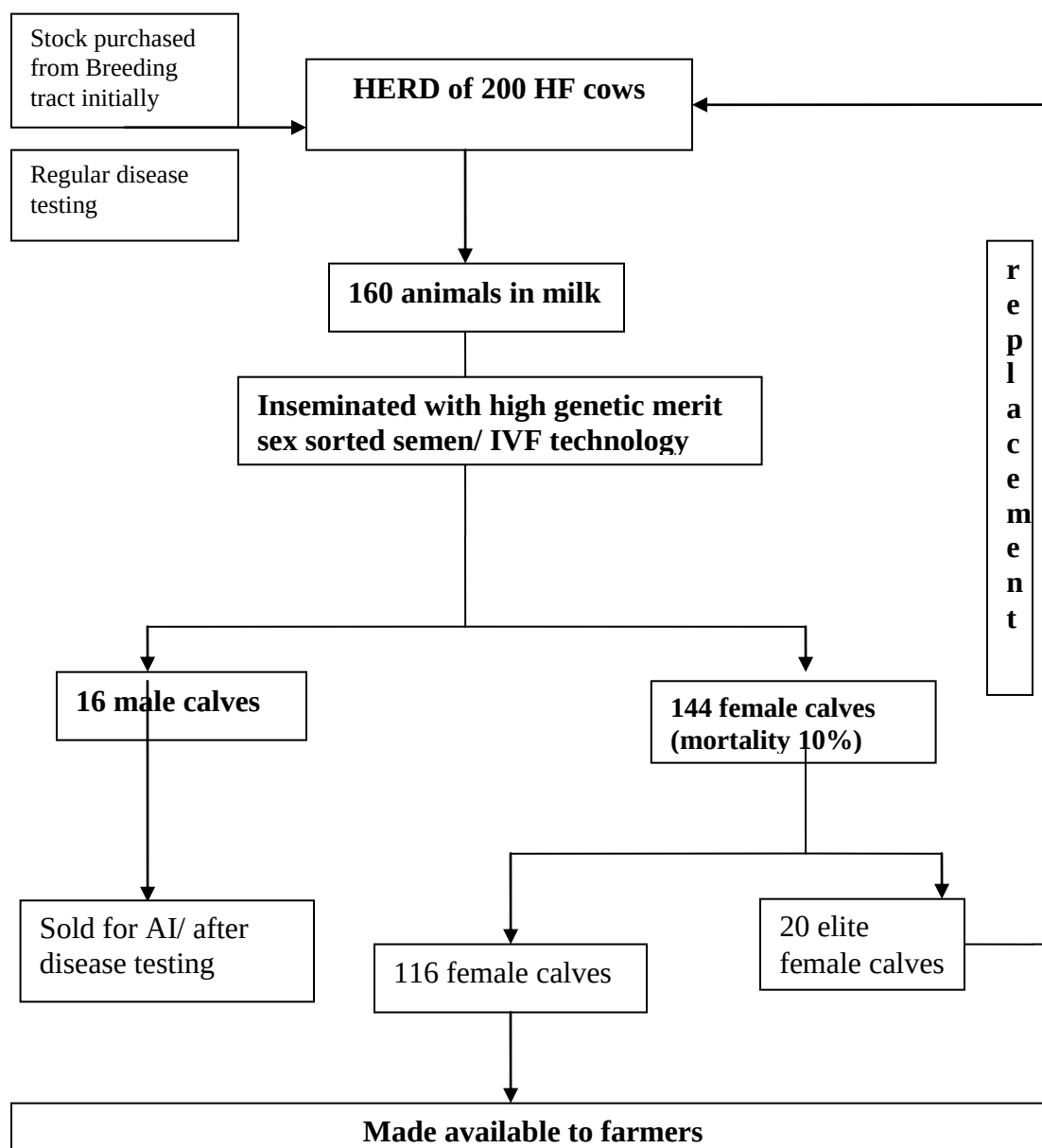


Fig: Technical Programme for BMF

SOURCING OF ANIMALS:

Pure bred animals of indigenous breeds will be procured from breeding tract. Animals will be selected on the basis of the dams performance (breeding value, lactation yield & milk fat %) and sires' information. Help of CHRS units be taken during procurement of animals.

CIVIL WORKS :

I Construction of cow sheds: 1 shed will be constructed with the capacity to house at least 300 animals. Shed will have all modern facilities. Fan and sprinklers will be fitted to prevent heat distress during summer months. Roof of the shed will be made from asbestos or other suitable material easily available in the area. Each animal will be provided with the floor space of 10 sq meter. Floor will be corrugated to avoid slipping. Adequate manger and water trough with access to drink water all time will be provided in all the sheds. Total cost of construction of sheds will be Rs 75 lakh . Modern cattle shed will be constructed having double row of cattle with head to head arrangement.

II Construction of calf pen: A shed with individual pens will be constructed to house calves born in the BMF. Each pen will have adequate water troughs and feeding space for the calves. Each animal will be provided with floor space of 3 sq meter. Total cost of construction of calf shed will be Rs 10 lakh

III Calving Pen: At one corner of BMF isolation shed will be constructed for separating ailing or sick animals for treatment. Sick animals are required to be removed from the herd for the safety of the animals. Isolation shed will be with pens to house sick animals. Facility will be created to house at least 10 animals.

IV. Establishment of feed store room:A store with the capacity to store 10 metric tones of feed and fodder will be established at BMF.

TRACTOR AND AGRICULTURE IMPLEMENTS

For fodder production and other associated activities at the Cattle Sanctuary, two tractor with agriculture implements will be required. For this an amount of Rs 14.00 lakh has been considered as capital investment.

Bio Gas Plant:

Bio gas plant of the capacity of 200 CU meter will be established at the farm and slurry and manure generated at the farm will made available to other farmers/ entrepreneurs on market rates.

Feed and Fodder arrangement:

Fodder crops will be grown at the farm including MP chari, Beseem, Lucerne, Maize, Oat, Barley, Jwar, etc. In addition dry fodder mainly wheat straw will be purchased from the farmers to meet requirement of the animals housed in the farm. Provision will be made at the farm for making silage from maize and other green suitable green fodder crops grown at the farm. Cattle feed ingredients will be purchased from the market and complete cattle feed will be prepared at the farm. Additional requirement of silage will be purchased from the known sources available in the country. Balanced ration will be fed to animals.

Water Arrangement for Animals

Tube well / any other source like pond, canal etc of water (give details) is already available at the farm. Water is of potable quality and sufficient water is available for drinking, cleaning of animals and cleaning of sheds and other utensils.

Veterinary aid:

Trevis has been constructed at the farm for treatment and conducting artificial insemination. Help of nearby local veterinarian will be taken for treatment of sick animals and also for artificial insemination. Local is available in the area and suitable arrangement has been made for this purpose.

RECURRING COST:

To maintain a dairy farm over a period of 5 years, revenue expenditure has been calculated which likely to be tune of Rs 1635.03lakh.

COST ESTIMATES:

The year wise financial details related to capital investment and recurring expenditure to be incurred under the project are given in Annexure-I and IB. An amount of R425 lakh would be required as capital investment under the project and Rs 1635.03 lakh will be needed as recurring expenditure for implementation of the project during 5 years.

OPERATING STATEMENT:

Revenue expenditure of Rs 1985.63 lakh and an income of Rs 2359.5 lakh have been estimated. Details are shown in Annexure- II.

Check List

- i) Please mention how you wish to establish and build the infrastructure on the proposed farm. (details given in the DPR)
- ii) Proposed Breed and number of animals planned to be reared on the farm. (Details given in DPR)

- iii) Rationale for choosing particular breed. (Details given in DPR)
- iv) Number and type of cattle sheds planned . (Details given in DPR)
- v) Expected milk production and number of calves to be produced every year (Details given in DPR)
- vi) Availability of water and related arrangements in the area and on the farm (Details given in DPR)
- vii) Plan for production and procurement of feed and fodder (Details given in DPR)
- viii)Arrangement for consultancy and veterinary aid (Details given in DPR)
- ix) Plan for dung management. (Details given in DPR)
- x) Plan for energy requirement (electricity and hot water supply)
- xi) Expected expenditure and income on the farm (Details given in DPR)
- xii) Mode of reproduction to be employed on the farm (Details given in DPR)
- xiii)Committed number of bull calves produced and heifers supplied to farmers from the farm (Details given in DPR)

Capital investment for Breed multiplication farm

Annexure-I

S. No	Particulars	Unit	Physical Targets					Unit cost	Financial Requirements					
								(Rs)	(Rs lakh)					
			1 st Year	2 nd Year	3 rd Year	4 th Year	5 th Year		1 st Year	2 nd Year	3 rd Year	4 th Year	5 th Year	Total
1	Purchase of cows in first lactation/2 nd lactation	No	200					100000	200					200
2	Construction of cow sheds 10 sq meter per cow (sheds to house 200 cows and its followers	per animal	300	100		---	---	25000	75					75
3	Calf pens (3 sq meter /calf)	Per animal	200	...			...	5000	10					10
4	Calving pens	per animal	10	-	-	-	-	25000	2.5					2.5
4	Administration block	Sq ft	1000	-	-	-	-	2000	20					20
5	Feed & fodder store room	sq ft	2000					2000	40					40

6	Tractor 75 HP, with agriculture implements (culti, seed drill, bailer, fodder harvester, mower, TMR wagon, other equipments as per requirement)	No	1					4000000	40					40
7	Dairy equipments (BMC, stainless steel Milk cans, digital milko tester, ghee making machine, deep freezer etc)	LS						1000000	10					10
8	Shed for Agri implements	sq ft	1500					1500	22.5					22.5
9	Chaff cutter (electric)		1					500000	5					5
10	Bio gas plant	Cu meter	1						10					
	Total								435	0	0	0	0	435

Breed multiplication farm
Recurring expenses

S. No	Particulars	Unit	Physical Targets					Unit cost(Rs)	Financial Targets (Rs lakh)					
			1 st Year	2 nd Year	3 rd Year	4 th Year	5 th Year		1 st Year	2 nd Year	3 rd Year	4 th Year	5 th Year	Total
	Variable Cost-A													
1	Feeding and maintenance	No	200	225	250	250	250	200/animal /day	146	164.25	182.5	182.5	182.5	857.75
2	Feeding of calves	No	160	160	200	200	200	100/calf /day	58.4	58.4	73	73	73	335.8
3	Feeding of heifers	No		160	160	160	160	150/heifer /day	0	87.6	87.6	87.6	87.6	350.4
4	Veterinary Medicine	No	100	100	100	100	100	500/year	1.8	0.545	0.61	0.61	0.61	4.175
5	Water & electricity charges	No	-	-	-	-	-	5000/month	0.6	0.6	0.6	0.6	0.6	3
6	Breeding charges	No	180	347	369	369	369	1000/animal	5.4	10.395	11.07	11.07	11.07	49.005
7	Milk chilling charges	Lakh lts	10.95	12.31875	13.6875	13.6875	13.6875	0.50/litre	5.475	6.159375	6.84375	6.84375	6.84375	32.166
8	Miscellaneous expenses	No	100	100	100	100	100	100/animal	0.36	0.545	0.61	0.61	0.61	2.735
	Total								218.04	328.4944	362.8338	362.8338	362.8338	1635.03

Breed multiplication farm
Recurring expenses on Manpower

Annexure-IB

S. No	Particulars	Unit	Physical Targets					Unit cost (Rs)	Financial Targets (Rs lakh)					
			1 st Year	2 nd Year	3 rd Year	4 th Year	5 th Year		1 st Year	2 nd Year	3 rd Year	4 th Year	5 th Year	Total
	Fixed Cost-B													
1	Manager	No	1					40000/month	4.8	4.8	4.8	0	9.6	24
2	Cow handlers/workers	No	10		5			15000/month	18	18	27	27	27	117
3	Chowkidar	No	2					15000/month	3.6	9	9	9	9	39.6
	Total								26.4	31.8	40.8	36	45.6	180.6

**Breed multiplication farm
Estimated Operating Cost & Income**

Annexure-II

S. No	Particulars	Financial (Rs lakh)					
		1 st Year	2 nd Year	3 rd Year	4 th Year	5 th Year	Total
	Expenditure						
1	Total Variable Cost A	218.035	328.494375	362.8338	362.8338	362.8338	1635.03
2	Fixed Cost B	26.4	31.8	40.8	36	45.6	180.6
3	Depreciation cost C @ 8% per year	34.8	34.8	34.8	34.8	34.8	174
	Total A+B +C	279.235	395.094	438.4338	433.633	443.2338	1989.63
	Income						
1	Income from sale of milk @ Rs 50/kg	360	360	360	360	360	1800
2	Income from sale of breeding stock @ Rs 1 lakh per animal @ 150 animals			150	150	150	450
3	Sale of organic manure Rs 2 per kg; 10 kg manure per cow per day	21.9	21.9	21.9	21.9	21.9	21.9
	Total income	381.9	381.9	531.9	531.9	531.9	2359.5
	Surplus/Deficit	102.66	-13.194	93.4662	98.26	88.66	369.87

Expected Cash Flow**Annexure-III**

S. No	Particulars	Financial (Rs lakh)					
		1 st Year	2 nd Year	3 rd Year	4 th Year	5 th Year	Total
1	Total non recurring expenditure	425	0	0	0	0	425
2	Total recurring expenditure on manpower	26.4	31.8	40.8	36	45.6	180.6
3	Total recurring on management of animals	218.035	328.49438	362.834	362.834	362.834	1635.03
	Total	669.435	360.294375	403.6338	398.8338	408.4338	2240.631

Guidelines for preparing project Document

A project proposal to establish a Breed Multiplication Farm for ----- Breed with ----- adult cows/buffaloes at-----.(to be filled by beneficiary)

Introduction:

[Should contain purpose of establishing farm, the location proposed, the breeds proposed (from Gir/Sahiwal/Red Sindhi/Rathi/Haryana/Tharparkar/Crossbred Jersey/Crossbred HF/Murrah/Mehsana/Nilli Ravi/Jaffarabadi),.....]

Dairying till date had been supplementary source of income to the farmers. With increasing population trend, frequent climate changes affecting productivity of agricultural produce, less opportunities in other industries in rural areas etc. has prompted many young farmers and entrepreneurs to take up dairy business in rural and semiurban areas. The scope for dairying has increased due to both increased population and increased income level leading to higher disposable income. This has led to increased demand for milk and dairy products in our country and all over the world.

Dairy business is environmentally successful if taken in scientific manner. Animal productivity affects sustainability and profitability of a dairy farm. In our country, there are many aspiring young farmers willing to take up dairy as a specialized profession. However, there are two main obstacle in this aspect. One is required technical knowhow for dairy farming and second one is availability of quality animals for dairy farmers.

Please also mention your area specific issues or other points you think are important for supporting your initiatives.

To tackle the issue of supply of quality animals I/we propose to establish a dairy farm with ---
- adult dairy cows of --- breed at -----.

Introduction of farmer

The farmer or the firm have to provide complete information on himself or his firm here

Name of the applicant:.....(to be filled by beneficiary)

Current occupation/current business and area of work:..... (to be filled by beneficiary)

Details of past experience of farm operations especially dairy farm operations/details on present arrangement with veterinary consultant or any such firm if any:

Purpose of establishment of the farm:

Please mention motivation that led you to decide to start a breed multiplication farm/ why you wish to establish this farm. Specifically describe gaps (availability of quality animals/training for farmers and entrepreneurs/supply of quality milk/milk products) that you wish to bridge by establishing this farm in your area.

Information on already available infrastructure:

Please describe current land holding, current dairy farm infrastructure, milk processing and marketing arrangements ect. Also describe current viability of the farm. You may put some photographs for existing infrastructure.

Information on proposed farm

Please mention how you wish to establish and build the infrastructure on the proposed farm.

- 1. Proposed Breed and number of animals planned to be reared on the farm. Rationale for choosing particular breed.*
- 2. Number and type of cattle sheds planned*
- 3. Expected milk production and number of calves to be produced every year*
- 4. Availability of water and related arrangements in the area and on the farm*
- 5. Plan for production and procurement of feed and fodder*
- 6. Arrangement for consultancy and veterinary aid*
- 7. Plan for dung management*
- 8. Plan for energy requirement (electricity and hot water supply)*
- 9. Expected expenditure and income on the farm*
- 10. Mode of reproduction to be employed on the farm*
- 11. Committed number of bull calves produced and heifers supplied to farmers from the farm.*

Please provide detailed statement on farm operation based on excel working.

Information on source of funding and required amount of subsidy based on scheme provisions:

Please provide details of estimates of capital investment requirement for next five years for various aspects:

Particular	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Civil infrastructure						
Equipment						
Animal purchase						
Total Capital funds required						
Funds to be met from own resources						
Funds to be sourced from bank loan						
Subsidy amount eligible						

Considering the requirement for supply of ----, I/we request DAHD to consider our proposal and grant a subsidy amount of Rs. ---. We assure that we will supply annually, ---- heifers till next 4 years or till --- 2026, to the interested farmers on cost basis. We also assure to abide by terms and conditions under which the proposed Breed Multiplication Farm will be funded under Rashtriya Gokul Mission.